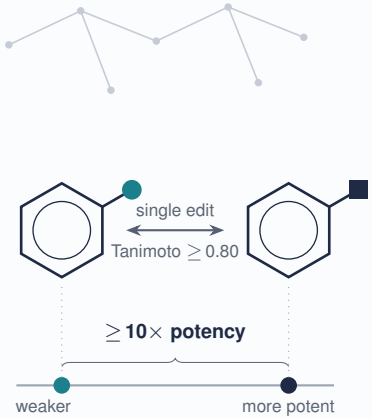
TELEMEDICON26 | PROJECT PITCH DECK

CliffForge

Relative-Affinity Molecular Analogue Prioritization and Activity-Cliff Analysis

Constrained analogue generation · relative-affinity scoring · uncertainty-aware ranking



an activity cliff (schematic)

Team Genesis

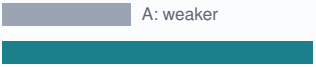
Chidwipak Kuppani – S20230010131 – UG4
Hrishikesh Kurapati – S20230010132 – UG4

Problem Statement & Validation

What is an activity cliff?

Structurally *similar* molecules whose measured potency differs *greatly*: a small edit, a large effect.





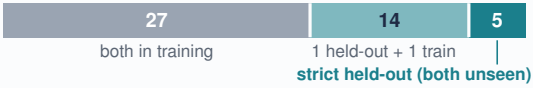
ΔpIC$_{50}$ $\geq$ 1.0 ($\geq$ 10$\times$ potency difference)

- **Why it matters:** chemists must choose which analogues to make; cliffs break "similar $\Rightarrow$ similar activity".
- Global absolute-activity models and similarity alone can rank close analogues poorly.

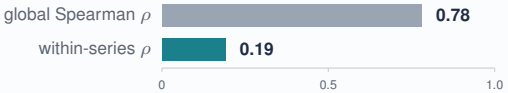
Validation on COVID Moonshot Mpro data

1,102 molecules · cliff = Tanimoto $\geq$ 0.80 **and** $|\Delta\text{pIC}_{50}| \geq 1.0$

46 cliff pairs in the full dataset



ChemProp (absolute), held-out Mpro compounds



Close held-out pairs: 44, containing 5 cliffs (12 test series).

Sources: (Maggiora 2006; Stumpfe & Bajorath 2012; van Tilborg et al. 2022); data (Boby et al. 2023); ChemProp (Yang et al. 2019); counts: this work.

Proposed Solution & Innovation

1 Constrained analogue generation

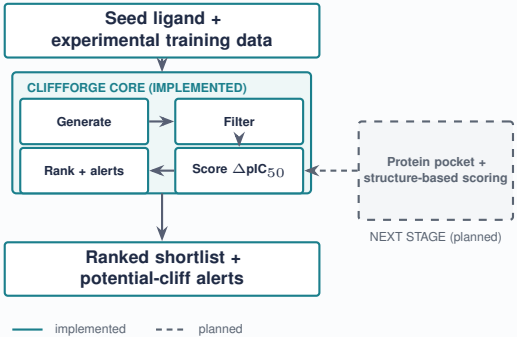
Training-derived substituent swaps (RDKit matched molecular pairs): small, traceable edits. Rule/example-based – not a neural generative model.

2 Relative-affinity scoring

Siamese D-MPNN trained on potency *differences* between related molecules; predicts candidate vs. seed ΔpIC_{50}.

3 Alerts & uncertainty-aware ranking

Three-model ensemble spread enters the ranking; potential-cliff flag for close analogues with a large predicted change.

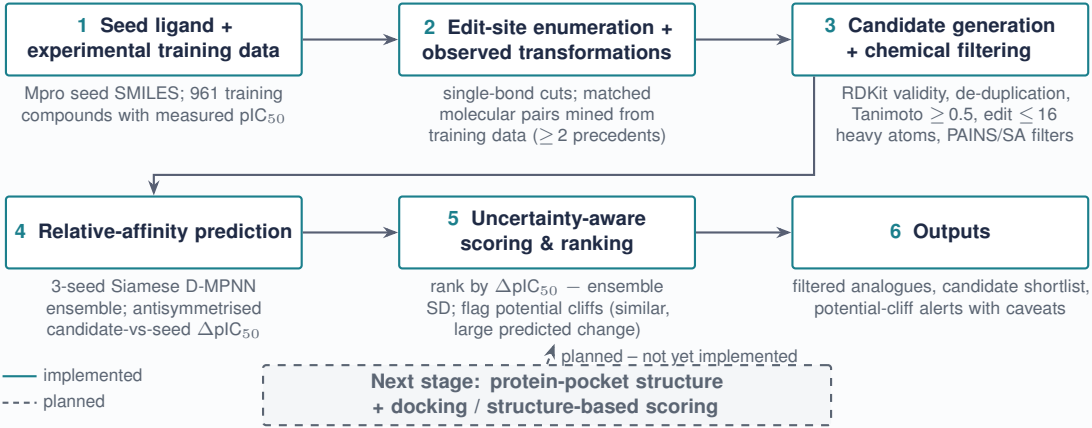


Technology stack: Python · RDKit (transformation transfer, filters) · PyTorch Siamese D-MPNN · experimental activity data for training & retrospective evaluation · next: protein-pocket / structure-based scoring

Intended users: drug-discovery research groups · medicinal chemists · pharmaceutical and biotechnology lead-optimization teams

Sources: matched molecular pairs (Griffen et al. 2011; Hussain & Rea 2010); Siamese networks (Bromley et al. 1993); PyTorch (Paszke et al. 2019); RDKit (rdkit.org).

Technical Methodology

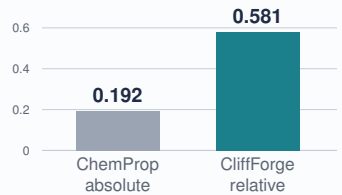


Sources: matched molecular pairs (Hussain & Rea 2010); D-MPNN as in ChemProp (Yang et al. 2019). Held-out activities are masked at inference.

Prototype / Proof of Concept

A Relative-affinity ranking

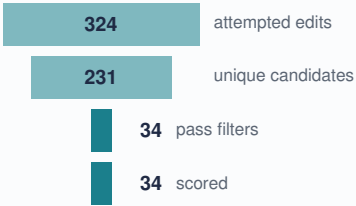
mean within-series Spearman ρ, held-out Mpro compounds



12 evaluation series; paired Wilcoxon $p = 0.005$.
Prior internship research (FEP+, 26 series): 0.386 → 0.644.

B Candidate-generation funnel

one held-out Mpro seed (pIC$_{50}$ 7.18)



≈ 27 s on CPU end-to-end (reference run).
Model-predicted scores; none synthesised or docked.

C Preliminary cliff detection

strict held-out pairs: 44 close pairs, 5 true cliffs

CliffForge relative



ChemProp baseline



Boxes = the 5 true cliffs; filled = flagged. No false-positive flags among the 44 pairs. Average precision **0.73** vs. **0.35**.

Preliminary: only 5 positive pairs.

Supporting retrospective check: training-derived edits recovered **82 of 316** reachable held-out analogue pairs; the relative model gave the correct activity direction on 53/82 (65%) vs. 47/82 (57%) for ChemProp (difference not significant).

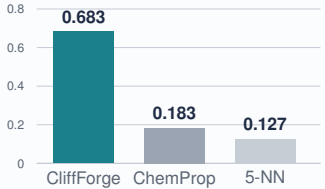
Sources: ChemProp (Yang et al. 2019); Moonshot data (Boby et al. 2023); FEP+ benchmark, prior work (Ross et al. 2023). All results are new except the FEP+ line. The lead-optimization setting (14 cliffs) gave mixed results and informs the next stage.

Validation at Scale: Dablander MMP Dataset (1,924 molecules)

Dataset: 1,924 Mpro molecules · 1,582 train / 181 test · 9,323 training MMP pairs
Cliff: Tanimoto ≥ 0.80 **and** $|\Delta \text{pIC}_{50}| \geq 1.0$

A Lead-optimisation cliffs

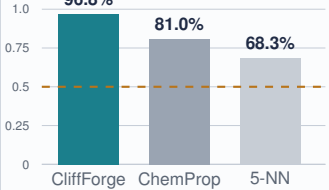
Average precision; Setting B (test vs. training)
25 true cliffs among 223 close pairs



Cliffs detected: 13/25 vs. 0/25 for both baselines

B Cliff-direction accuracy

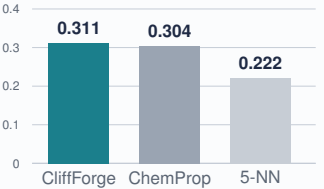
63 test-test pairs with $|\Delta \text{pIC}_{50}| \geq 1.0$
Dashed line = 50% chance



CliffForge: +16 pp vs. ChemProp

C Within-series ranking

Mean Spearman ρ
12 held-out series



CliffForge: 0.311 vs. ChemProp: 0.304

Key finding: In Setting B, CliffForge detects 13/25 cliffs (F1 = 0.634, precision = 81%); ChemProp and 5-NN detect none (F1 = 0). Direction accuracy: 96.8% vs. ChemProp 81.0% (**+16 pp**). Setting A (2 strict cliffs): AP 1.00 vs. 0.08 (CliffForge vs. ChemProp).

Data: (Boby et al. 2023; Dablander et al. 2023); 1,924 molecules, 12,594 MMPs; results in `results/summary_mmp.json`. ChemProp 2.x (D-MPNN depth=3, 60 epochs). All evaluations are strictly held-out.

Impact, Sustainability & Next Steps

Potential impact (if further validated)

More efficient analogue prioritization, with explicit uncertainty, to guide follow-up experiments.

Prospective users (not confirmed partners)

- Academic drug-discovery groups and medicinal chemistry teams
- Biotechnology start-ups and pharmaceutical R&D
- Research collaborations, possibly CRO partners

Possible sustainability models (proposals)

- Open research toolkit
- Research collaborations
- Optional hosted workflows / institutional integration

Next-stage roadmap

✓ **Today: working prototype**
generation, relative scoring, ranking, evaluation tooling

1. **Expanded cliff validation**
larger, series-rich datasets; more held-out positive pairs
2. **Structure-based scoring**
suitable protein structure + docking workflow
3. **Stereochemistry- and 3D-aware evaluation**
beyond the current 2D, stereo-free models
4. **Richer transformation library**
expand and measure candidate coverage
5. **Experimental test**
selected candidate set, when feasible

Value: relative-affinity learning and activity-cliff-informed analogue prioritization aim to focus effort on the most informative candidates.